

Hyunjin Lim

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EDUCATION

Seoul National University

Seoul, South Korea

B.S. in Computer Science and Engineering (incl. mandatory military service)

Mar. 2025 – Feb. 2029 (expected)

- GPA: 4.15/4.30 (Major GPA: 4.30/4.30)
- Merit-based Academic Excellence Scholarship (top 15% of department), Fall 2025

Daegu Science High School for the Gifted

Daegu, South Korea

1 of 8 national high schools for South Korea's top STEM students

Mar. 2021 – Feb. 2024

- **Silver Medal**, In-School Mathematics Competition (2022)

COURSEWORK HIGHLIGHTS

Mathematics: Real Analysis, Probability Theory, Linear Algebra, Discrete Mathematics, Calculus I & II

Computer Science: Data Structures, Computer Programming

PROJECTS

Degree Reduction for QUBO on Graph Coloring | *Combinatorics, Quantum Annealing*

Mar. – Dec. 2022

- Year-long team R&E project at Daegu Science High School for the Gifted, co-mentored by a DGIST faculty member; studied degree reduction of high-order polynomials in QUBO formulations of the graph coloring problem for quantum annealing (D-Wave)
- Identified a combinatorial pattern in raw numerical outputs through hand-analysis of computational examples; conjectured a closed-form identity for homogeneous symmetric polynomials on binary variables expressible via binomial coefficients
- Co-derived a rigorous combinatorial proof that generalizes the classical monomial reduction methods of Freedman and Ishikawa; the resulting algorithm yields reduced quadratic polynomials with fewer auxiliary variables than the conventional methods
- Co-authored [arXiv:2306.12081](https://arxiv.org/abs/2306.12081); awarded **Best R&E Research Prize** (1st place in mathematics, school-wide)

LEADERSHIP & ACTIVITIES

Student Council, College of Engineering

Seoul National University

Chair of Executive Committee; Deputy Head, Education Affairs Division

Dec. 2025 – Present

- Lead a 70-member executive committee overseeing student council operations across the College of Engineering, with an annual budget of 300M+ KRW (\$220K+)
- Designed and executed a college-wide student survey (N=250); used the quantitative findings as the negotiating basis with the College's dean's office on educational policy issues
- Mediated a contested decision with the administrative office on a joint-college festival by laying out the trade-offs as explicit scenarios; led the committee to converge on the highest-expected-value option

TECHNICAL SKILLS

Programming Languages: Python, C++

Languages: Korean (native), English (fluent)

INTERESTS

Strategy Games: Chess, Poker (Texas Hold'em)

Other: Basketball, Piano